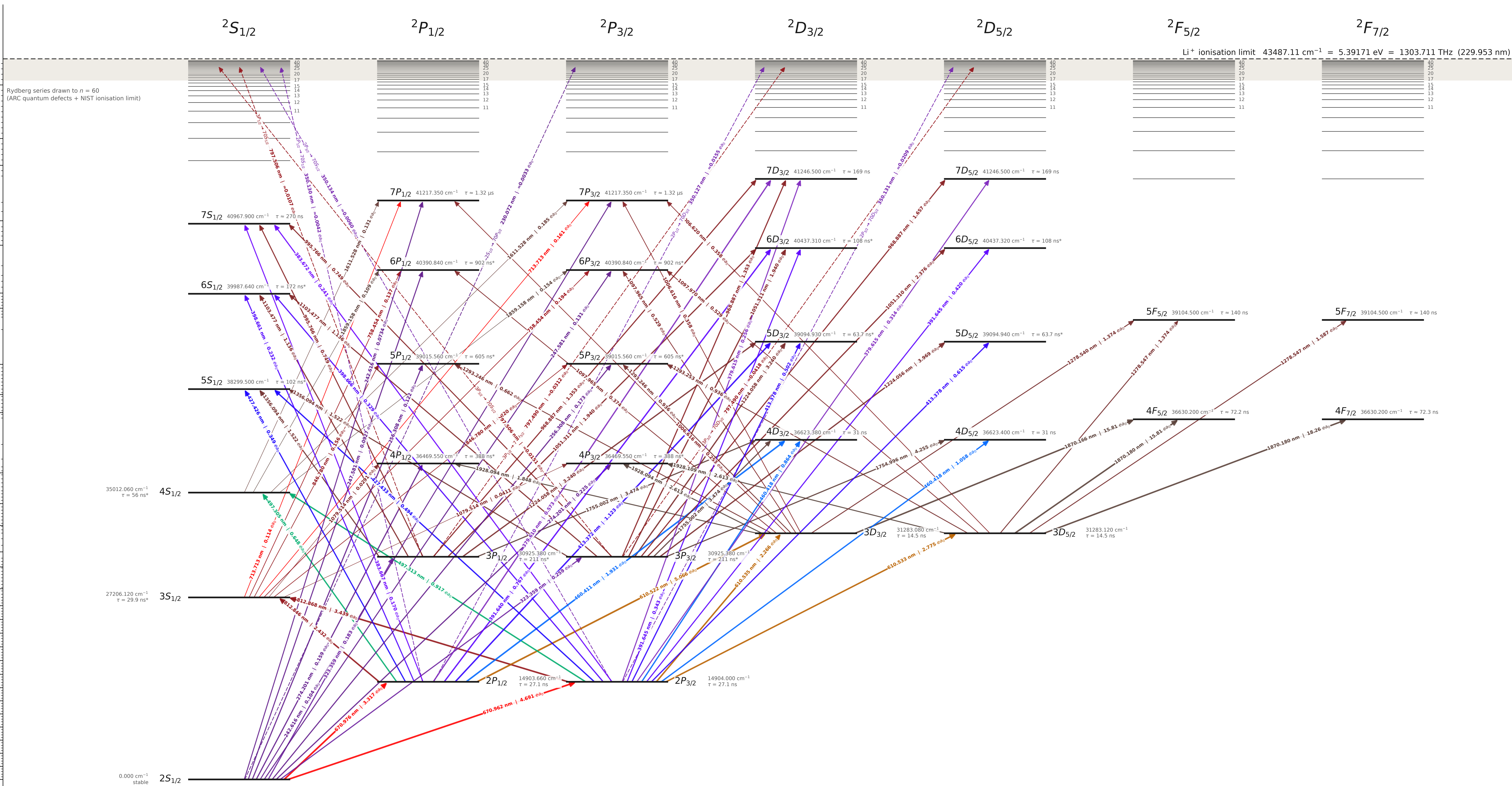


⁷Li (Li I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm: 32 levels, 85 lines.

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Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\bar{\nu}$ (cm ⁻¹)	$ \langle J e J' \rangle $ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Typical use
$2P_{1/2} - 3D_{3/2}$	610.5222	610.3533	491.0427	16379.420	5.066	5.714e+07	10.980	83.3	
$2S_{1/2} - 2P_{3/2}$	670.9616	670.7764	446.8102	14903.983	4.691	3.690e+07	5.872	100.0	D2 line
$2S_{1/2} - 2P_{1/2}$	670.9761	670.7909	446.8005	14903.660	3.317	3.690e+07	5.872	100.0	D1 line
$2P_{3/2} - 3S_{1/2}$	812.8680	812.6446	368.8083	12302.120	3.439	2.231e+07	5.329	66.7	
$2P_{1/2} - 4D_{3/2}$	460.4111	460.2821	651.1408	21719.720	1.931	1.935e+07	5.134	64.5	
$2P_{3/2} - 3D_{5/2}$	610.5334	610.3644	491.0337	16379.120	2.775	1.143e+07	10.980	100.0	
$2P_{3/2} - 3D_{3/2}$	610.5349	610.3659	491.0325	16379.080	2.266	1.143e+07	10.980	16.7	
$2P_{1/2} - 3S_{1/2}$	812.8456	812.6221	368.8185	12302.460	2.432	1.116e+07	5.329	33.3	
$2P_{1/2} - 5D_{3/2}$	413.3723	413.2557	725.2360	24191.270	1.123	9.040e+06	2.498	57.5	
$2P_{3/2} - 4S_{1/2}$	497.3130	497.1743	602.8245	20108.060	0.917	6.919e+06	2.840	38.8	
$2P_{1/2} - 6D_{3/2}$	391.6401	391.5292	765.4796	25533.650	0.767	4.964e+06	1.469	53.9	
$2P_{3/2} - 4D_{5/2}$	460.4179	460.2889	651.1312	21719.400	1.058	3.871e+06	5.134	77.4	
$2P_{3/2} - 4D_{3/2}$	460.4183	460.2893	651.1306	21719.380	0.864	3.871e+06	5.134	12.9	
$2P_{1/2} - 4S_{1/2}$	497.3046	497.1659	602.8347	20108.400	0.648	3.460e+06	2.840	19.4	
$2P_{3/2} - 5S_{1/2}$	427.4326	427.3124	701.3795	23395.500	0.494	3.170e+06	1.560	32.0	
$2P_{1/2} - 7D_{3/2}$	379.6098	379.5020	789.7385	26342.840	0.573	3.041e+06	0.941	52.2	
$2P_{3/2} - 5D_{5/2}$	413.3779	413.2613	725.2261	24190.940	0.615	1.810e+06	2.498	69.0	

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ_{air} uses standard air (Ciddor).
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$. Branch: share of the upper level's decays that takes this line.

Lydberg excitation wavelengths (vacuum, nm)							
Lower $\rightarrow$ series	n = 30	n = 40	n = 50	n = 70	n = 100	n $\rightarrow$ ∞	d_{70} ARC
$2S_{1/2} \rightarrow nP_{3/2}$	230.602	230.317	230.186	230.072	230.011	229.953	0.0033
$2P_{3/2} \rightarrow nS_{1/2}$	351.397	350.715	350.404	350.134	349.992	349.857	0.0060
$2P_{3/2} \rightarrow nD_{5/2}$	351.356	350.698	350.395	350.131	349.991	349.857	0.0209
$3S_{1/2} \rightarrow nP_{3/2}$	618.862	616.817	615.877	615.060	614.628	614.213	0.0052
$3P_{3/2} \rightarrow nS_{1/2}$	804.085	800.528	798.905	797.506	796.770	796.068	0.0151
$3P_{3/2} \rightarrow nD_{5/2}$	803.872	800.439	798.860	797.490	796.764	796.068	0.0418
$3S_{1/2} \rightarrow nP_{3/2}$	827.699	824.406	822.367	820.912	820.142	819.404	0.0077
$3D_{5/2} \rightarrow nF_{7/2}$	827.672	824.035	822.361	820.910	820.141	819.404	0.0449
$4S_{1/2} \rightarrow nP_{3/2}$	1197.211	1189.582	1186.088	1183.064	1181.465	1179.933	0.0070
$4P_{3/2} \rightarrow nS_{1/2}$	1450.888	1439.347	1434.111	1429.610	1427.245	1424.996	0.0276
$4P_{3/2} \rightarrow nD_{5/2}$	1450.195	1439.061	1433.965	1429.558	1427.228	1424.996	0.0670
$4D_{5/2} \rightarrow nF_{7/2}$	1483.369	1471.676	1466.332	1461.713	1459.272	1456.937	0.0184
$4D_{5/2} \rightarrow nF_{7/2}$	1483.285	1471.641	1466.314	1461.706	1459.270	1456.937	0.0837
$4F_{7/2} \rightarrow nD_{5/2}$	1484.786	1473.117	1467.778	1463.161	1460.720	1458.382	0.0080
$5S_{1/2} \rightarrow nP_{3/2}$	1974.213	1953.555	1944.136	1936.037	1931.758	1927.668	0.0086

Rydberg term energies from ARC quantum defects + NIST ionisation limit, lower level from NIST. $d_{70} = |\langle j||er||j \rangle|$ to $n = 70$ in ea₀ (calculated). $S \rightarrow nP$ values come from a near-cancelling integral: treat them as an order of magnitude.

Li hyperfine structure ($I = 3/2$)			
Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$2S_{1/2}$	401.752043	-	F=1: -502.19 F=2: +301.31
$2P_{1/2}$	46.005(16)	-	F=1: -57.51 F=2: +34.50
$2P_{3/2}$	-3.050(16)	-0.22(3)	F=0: +11.16 F=1: +8.33 F=2: +2.45 F=3: -6.92
$3S_{1/2}$	93.105(7)	-	F=1: -116.38 F=2: +69.83
$3P_{1/2}$	13.50(20)	-	F=1: -16.88 F=2: +10.12
$3P_{3/2}$	-1.010(20)	-0.081(28)	F=0: +3.69 F=1: +2.76 F=2: +0.82 F=3: -2.29
$3D_{3/2}$	0.84(4)	-	F=0: -3.16 F=1: -2.32 F=2: -0.63 F=3: +1.90
$3D_{5/2}$	0.3436(10)	-	F=1: -1.80 F=2: -1.12 F=3: -0.09 F=4: +1.29
$4S_{1/2}$	35.0(4)	-	F=1: -43.81 F=2: +26.29
$4P_{3/2}$	-0.41(4)	-	F=0: +1.54 F=1: +1.13 F=2: +0.31 F=3: -0.92

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).